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Unit – 1 : Session – 12 : SLO – 1 | Logic gates (combined)

I. Match the Following

1. Symbol of AND gate — standard AND symbol (■), $Y = A \text{ ■ } B$
2. Output of OR gate when $A=0$, $B=1$ — 1
3. NOT gate logic function — c) $Y = A \text{ ■}$
4. NOR gate implementation — d) Universal gate to implement AND, OR, NOT
5. EX-OR practical use — e) Parity generator/checker, Half Adder

II. Fill in the Blanks

1. all 2. any 3. 0 4. universal 5. different

III. Activity / Describe and Perform

1. AND/OR/NOT — AND: $Y = A \text{ ■ } B$, truth table $00 \rightarrow 0$, $01 \rightarrow 0$, $10 \rightarrow 0$, $11 \rightarrow 1$. OR: $Y = A + B$, $00 \rightarrow 0$, $01 \rightarrow 1$, $10 \rightarrow 1$, $11 \rightarrow 1$. NOT: $Y = A \text{ ■}$, $0 \rightarrow 1$, $1 \rightarrow 0$. Symbols: D-shape (AND), curved (OR), triangle+bubble (NOT).
2. NOR gate: $Y = (A + B) \text{ ■}$ — output 1 only when both inputs are 0. Truth table: $00 \rightarrow 1$, $01 \rightarrow 0$, $10 \rightarrow 0$, $11 \rightarrow 0$. Implementations: NOT = $A \text{ NOR } A$; OR = $(A \text{ NOR } B) \text{ NOR } (A \text{ NOR } B)$; AND = $(A \text{ NOR } A) \text{ NOR } (B \text{ NOR } B)$.
3. EX-OR gate: $Y = A \oplus B = A \text{ ■ } B + AB \text{ ■}$ — output 1 when inputs differ. Truth table: $00 \rightarrow 0$, $01 \rightarrow 1$, $10 \rightarrow 1$, $11 \rightarrow 0$. Uses: parity generator/checker, Half Adder sum output, digital comparators.